

Lattice-Based Constant Deformation Jacobians for Shape Control of Deformable Objects

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Abstract—In this article, we propose a novel approach for representing the deformation behavior of an elastic object. The approach works by fitting a uniform lattice to the object and defining an as-similar-as-possible deformation energy for every set of three consecutive aligned nodes along the principal axes of this lattice. Based on a geometric analysis for each such set, we derive a quasi-static constant deformation Jacobian for the full lattice. An interesting application of this Jacobian is in shape control of deformable objects via robotic manipulation, as it removes the need for mechanical object characterization and online model adaptation. While the use of the proposed Jacobian model for control relies on the assumption of small local deformations, the model can still be used to control significant global deformations. In addition, it can represent deformable objects of diverse forms (e.g., beam-like or sheet-like). We validate this model through shape control simulations.

I. INTRODUCTION

The automatic manipulation of deformable objects finds many relevant applications in industrial, domestic, agricultural or medical domains [1], [2]. When a manipulation task involves a robot having to control the shape of a deformable object, a common approach is to use a mechanical model of the object in order to predict its deformation [3]–[5]. For specific geometries such as linear structures, well-known frameworks such as Kirchhoff or Cosserat rods have been employed [6]–[8]. Mechanical models can be successfully used both for planning and for online control based on sensor feedback. However, a common challenge in practice is the unavailability of a mechanical characterization of the object.

To address this challenge, some authors resorted to adaptive estimation techniques where a deformation Jacobian (i.e., a linearized mapping from robot motion to object deformation) is obtained based on sensor data [9], [10]. Learned models have also been popular. In [11], a multilayer neural network was exploited to model the deformation of objects of different types. The work in [12] demonstrated a cable installation task leveraging a graph neural network. Offline learning and online adaptation of a radial-basis-function neural network were used for shape control in [13].

Other approaches attempt to approximate the deformation behavior by making certain geometric assumptions. A usual

assumption of this type is diminishing rigidity [14]. The assumption in the work [15] is that the object stays rigid within an adjustable local region. A different type of strategy is to use geometric models from the domain of computer graphics. An example is the popular model called as-rigid-as-possible [16], which encodes a general tendency to preserve local rigidity and has been employed, e.g., in [17] to control the shape of diverse objects. While these geometric models are not meant to be a physically accurate representation of the specific object being manipulated, they are still accurate enough for quasi-static shape control purposes when used alongside corrective sensor feedback. In shape control tasks, the main advantages of geometric models are that they are simple to implement and fast to compute, and they avoid the need for mechanical characterization.

Lattice-based geometric models, in particular, simplify the handling of deformation by embedding the object's geometry in a uniform lattice and modeling the deformation of the lattice only [17]–[19]. In this work, we propose a new lattice-based geometric model of deformation aimed at shape control. We propose a grouping of the nodes in a 2D or 3D uniform lattice into sets of three consecutive nodes along its principal axes, and formulate a deformation energy for each such set based on an as-similar-as-possible assumption. We show that the energy for every set is a positive semidefinite quadratic form in the node positions, which allows us to derive a quasi-static constant Jacobian mapping between displacements of gripped nodes and non-gripped nodes of the lattice. Having a constant Jacobian is highly relevant for shape control, as it avoids the need for adaptive estimations and only requires sensing the state of the controlled parts on the object, and not its full shape.

Various prior works employed as-similar-as-possible formulations. Some of these works focused on objects restricted to 2D workspaces [20], [21]. The work in [22] considered deformations in 3D and dealt with shape control, considering only one-dimensional lattices and providing no formal results. By comparison, our approach accommodates more general objects and deformations, and is grounded in theoretical analysis. In [23], an as-similar-as-possible error functional was proposed, formulated as an approximation that is valid for small-angle rotations; in contrast, thanks to the particular grouping of nodes we employ, our as-similar-as-possible formulation does not involve any approximations.

The formulation we present is mainly aimed at being used for shape control of elastic objects. Elastic objects exhibit as-rigid-as-possible deformations [16], [19], associated with local rigid transformations. They do not exhibit the as-

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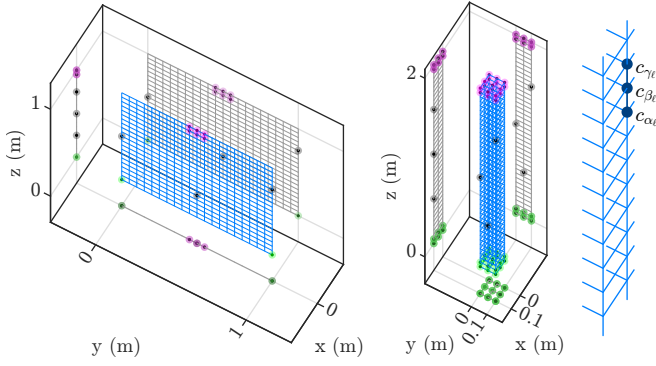


Fig. 1. 2D (left) and 3D (center) lattices. The dots represent the anchored (green), gripped (pink) and servoed (black) nodes, discussed in Section III. Positions of a generic three-node set $\mathcal{T}_\ell$ are shown on the right.

similar-as-possible deformations, associated with similarity transformations, that we assume when computing the Jacobian. However, for an elastic object under small local deformations, the scaling of the local similarity transformation remains close to one [21]. Moreover, the global deformation of the object can still be substantial even when the deformation is always small in every local region. For these reasons, it can be feasible to use the proposed Jacobian to control large deformations of elastic objects. We support this fact via shape control tests with as-rigid-as-possible-modeled virtual objects, a Model Predictive Control (MPC) based on the proposed Jacobian and a Monte Carlo analysis. The simulations yield promising results for shape control under significant deformations.

II. LATTICE MODEL

Consider a deformable object represented by a uniform square (in 2D) or cubical (in 3D) lattice of n nodes. To illustrate this, Fig. 1 shows two lattice configurations: the left one is a 2D lattice of a $1 \text{ m} \times 1 \text{ m}$ square, with 441 nodes; the one in the center is a 3D lattice of a $0.1 \text{ m} \times 0.1 \text{ m} \times 2 \text{ m}$ prism, with 369 nodes. The lattice is defined based on the geometry of the object at rest, in such a way that the lattice fully contains the geometry of the object. The object is modeled as bound geometrically to the lattice, and hence it deforms following the deformation of the lattice. In this way, lattice-based approaches allow for efficient and versatile handling of deformation [17]–[19].

The positions of the nodes of the lattice at rest are denoted by $\mathbf{c}_\kappa \in \mathbb{R}^3$, for $\kappa = 1, 2, \dots, n$. The positions of the nodes of the lattice in its current configuration are denoted by $\mathbf{p}_\kappa \in \mathbb{R}^3$, for $\kappa = 1, 2, \dots, n$. All these positions are expressed in an arbitrary coordinate frame. Let $\mathbf{p} = [\mathbf{p}_1^\top, \mathbf{p}_2^\top, \dots, \mathbf{p}_n^\top]^\top \in \mathbb{R}^{3n}$ denote the stack vector encoding the current lattice shape.

A. Deformation Energy

The model we propose for representing the deformation behavior is based on grouping the lattice's nodes in overlapping local regions. Specifically, every three neighboring nodes along every principal axis of the lattice are grouped

in a set that represents a local region. In order to make this grouping possible, we assume there are at least three nodes along every axis of the lattice. Let us denote by m the total number of three-node sets and assign an index to each of them. Concretely, we denote the ℓ -th set by $\mathcal{T}_\ell = \{\alpha_\ell, \beta_\ell, \gamma_\ell\}$, where $\alpha_\ell, \beta_\ell, \gamma_\ell$ are in $\{1, 2, \dots, n\}$ and ℓ in $\{1, 2, \dots, m\}$. Here, β_ℓ is defined, without loss of generality, as the index of the node being at the center position; i.e., the points $\mathbf{c}_{\alpha_\ell}, \mathbf{c}_{\beta_\ell}, \mathbf{c}_{\gamma_\ell}$ form a straight line, with $\mathbf{c}_{\beta_\ell}$ at the center. An illustration appears in Fig. 1.

We define a deformation energy $E \in \mathbb{R}_{\geq 0}$ for the full lattice as a sum of an energy $E_\ell \in \mathbb{R}_{\geq 0}$ for every local region:

$$E = \sum_{\ell=1}^m E_\ell. \quad (1)$$

This is understood as a potential energy associated with the lattice's deformation. When the lattice is in a deformed state, this energy creates internal forces on the nodes, pushing every local region toward its undeformed state. We model every E_ℓ as an as-similar-as-possible (ASAP) energy. This means the rest state for set $\mathcal{T}_\ell$ is represented by the three points $\mathbf{c}_{\alpha_\ell}, \mathbf{c}_{\beta_\ell}, \mathbf{c}_{\gamma_\ell}$, and the undeformed state is represented by three points equal to those of the rest state up to a geometric similarity (i.e., rotation, translation, and scaling). Hence, we define

$$E_\ell = \frac{1}{2} \sum_{\kappa \in \mathcal{T}_\ell} \|s_\ell \mathbf{R}_\ell \mathbf{c}_\kappa + \mathbf{t}_\ell - \mathbf{p}_\kappa\|^2 \quad (2)$$

for $\ell = 1, 2, \dots, m$, where $\|\cdot\|$ denotes the Euclidean norm. Concretely, the values of $s_\ell, \mathbf{R}_\ell$, and $\mathbf{t}_\ell$ in (2) are solutions to the problem

$$\arg \min_{s_\ell \in \mathbb{R}_{>0}, \mathbf{R}_\ell \in SO(3), \mathbf{t}_\ell \in \mathbb{R}^3} \sum_{\kappa \in \mathcal{T}_\ell} \|s_\ell \mathbf{R}_\ell \mathbf{c}_\kappa + \mathbf{t}_\ell - \mathbf{p}_\kappa\|^2, \quad (3)$$

for $\ell = 1, 2, \dots, m$. ASAP modeling and as-rigid-as-possible (ARAP) modeling differ in the scaling factor s_ℓ , which is set as equal to 1 in the latter. For small local deformations, s_ℓ in ASAP remains close to one. This motivates us to use the simpler ASAP model to approximate ARAP [21] and, therefore, to represent deformation behaviors of elastic objects. More details are given in Section III.

Let $\otimes$ denote the Kronecker product, $\mathbf{I}_k$ denote the $k \times k$ identity matrix, and $\mathbf{M}[i, j]$ denote the entry at row i and column j of matrix $\mathbf{M}$. Next, we present our derivation of a compact expression of the energy E .

Theorem 1: The ASAP energy E in (1) for a shape $\mathbf{p}$ of the lattice, where E_ℓ are as in (2) with $s_\ell, \mathbf{R}_\ell$ and $\mathbf{t}_\ell$ being solutions of (3), can be expressed as follows:

$$E = \frac{1}{2} \mathbf{p}^\top \mathbf{A} \mathbf{p}, \quad (4)$$

where the matrix $\mathbf{A} \in \mathbb{R}^{3n \times 3n}$ is

$$\mathbf{A} = \left(\sum_{\ell=1}^m \mathbf{Q}_\ell^\top \mathbf{B} \mathbf{Q}_\ell \right) \otimes \mathbf{I}_3, \quad (5)$$

with the matrix $\mathbf{B} \in \mathbb{R}^{3 \times 3}$ being

$$\mathbf{B} = \begin{bmatrix} 1/6 & -1/3 & 1/6 \\ -1/3 & 2/3 & -1/3 \\ 1/6 & -1/3 & 1/6 \end{bmatrix} \quad (6)$$

and $\mathbf{Q}_\ell \in \mathbb{R}^{3 \times n}$ for $\ell = 1, 2, \dots, m$ being selector matrices which satisfy $\mathbf{Q}_\ell[1, \alpha_\ell] = 1$, $\mathbf{Q}_\ell[2, \beta_\ell] = 1$, $\mathbf{Q}_\ell[3, \gamma_\ell] = 1$, and $\mathbf{Q}_\ell[i, j] = 0$ for any other pair of i, j .

Proof: We start by finding an expression for E_ℓ in (2), which requires choosing s_ℓ , $\mathbf{R}_\ell$, and $\mathbf{t}_\ell$ as solutions to the problem (3). Since the translation $\mathbf{t}_\ell$ is an isolated addend, the $\mathbf{t}_\ell$ that minimizes the quadratic cost in (3) can be found as in [24]: one assumes that $\mathbf{R}_\ell$ and s_ℓ are fixed and that $\mathbf{t}_\ell$ can vary, differentiates the cost with respect to $\mathbf{t}_\ell$, and finds the roots. This gives

$$\sum_{\kappa \in \mathcal{T}_\ell} (s_\ell \mathbf{R}_\ell \mathbf{c}_\kappa + \mathbf{t}_\ell - \mathbf{p}_\kappa) = \mathbf{0}. \quad (7)$$

Now, defining the centroids of the sets of three points $\{\mathbf{p}_\kappa\}_{\kappa \in \mathcal{T}_\ell}$ and $\{\mathbf{c}_\kappa\}_{\kappa \in \mathcal{T}_\ell}$ for the three-node set $\mathcal{T}_\ell$ as

$$\bar{\mathbf{p}}_\ell = \frac{1}{3} \sum_{\kappa \in \mathcal{T}_\ell} \mathbf{p}_\kappa, \quad \bar{\mathbf{c}}_\ell = \frac{1}{3} \sum_{\kappa \in \mathcal{T}_\ell} \mathbf{c}_\kappa, \quad (8)$$

one finds by solving (7) that the optimal translation is

$$\mathbf{t}_\ell = \bar{\mathbf{p}}_\ell - s_\ell \mathbf{R}_\ell \bar{\mathbf{c}}_\ell. \quad (9)$$

Defining now for each $\mathcal{T}_\ell$ the variables

$$\mathbf{p}'_\kappa = \mathbf{p}_\kappa - \bar{\mathbf{p}}_\ell, \quad \mathbf{c}'_\kappa = \mathbf{c}_\kappa - \bar{\mathbf{c}}_\ell, \quad \kappa \in \mathcal{T}_\ell, \quad (10)$$

notice that the problem (3) can be simplified into

$$\arg \min_{s_\ell \in \mathbb{R}_{>0}, \mathbf{R}_\ell \in SO(3)} \sum_{\kappa \in \mathcal{T}_\ell} \|s_\ell \mathbf{R}_\ell \mathbf{c}'_\kappa - \mathbf{p}'_\kappa\|^2, \quad (11)$$

for $\ell = 1, 2, \dots, m$.

Now, we exploit the specific geometric constraints of our scenario, which will allow us to obtain an analytic solution. In particular, the three points $\mathbf{c}_{\alpha_\ell}$, $\mathbf{c}_{\beta_\ell}$, $\mathbf{c}_{\gamma_\ell}$ in the shape at rest are consecutive along a principal axis of the lattice, they are evenly spaced, and they form a straight line. Hence, the node β_ℓ is located exactly at the middle point between the nodes α_ℓ and γ_ℓ in the shape at rest, and as a result we have the following two constraints:

$$\mathbf{c}'_{\beta_\ell} = \mathbf{0}, \quad \mathbf{c}'_{\gamma_\ell} = -\mathbf{c}'_{\alpha_\ell}. \quad (12)$$

Developing the addends in the cost of (11) and applying the constraints (12), that cost has the following form:

$$\|\mathbf{p}'_{\alpha_\ell}\|^2 + \|\mathbf{p}'_{\beta_\ell}\|^2 + \|\mathbf{p}'_{\gamma_\ell}\|^2 + 2s_\ell^2 \|\mathbf{c}'_{\alpha_\ell}\|^2 - 2s_\ell (\mathbf{R}_\ell \mathbf{c}'_{\alpha_\ell})^\top (\mathbf{p}'_{\alpha_\ell} - \mathbf{p}'_{\gamma_\ell}). \quad (13)$$

The last addend in this expression can be developed as

$$-2s_\ell \|\mathbf{c}'_{\alpha_\ell}\| \|\mathbf{p}'_{\alpha_\ell} - \mathbf{p}'_{\gamma_\ell}\| \cos(\theta_\ell), \quad (14)$$

where θ_ℓ is the angle formed by the two vectors $\mathbf{R}_\ell \mathbf{c}'_{\alpha_\ell}$ and $\mathbf{p}'_{\alpha_\ell} - \mathbf{p}'_{\gamma_\ell}$. Hence, minimizing the cost in (11) requires maximizing $\cos(\theta_\ell)$, i.e., making $\theta_\ell = 0$. Therefore, $\mathbf{R}_\ell$ is

chosen so that the two vectors are parallel and point in the same direction. This implies that the following relation holds:

$$\mathbf{R}_\ell \mathbf{c}'_{\alpha_\ell} = \frac{\|\mathbf{c}'_{\alpha_\ell}\|}{\|\mathbf{p}'_{\alpha_\ell} - \mathbf{p}'_{\gamma_\ell}\|} \cdot (\mathbf{p}'_{\alpha_\ell} - \mathbf{p}'_{\gamma_\ell}). \quad (15)$$

The condition $\mathbf{p}'_{\alpha_\ell} - \mathbf{p}'_{\gamma_\ell} = \mathbf{0}$ would correspond to cases where the nodes α_ℓ and γ_ℓ occupy the same position. Such cases are infeasible for real objects and can be disregarded. Hence, using the expression (15), we can write (13) as follows:

$$\|\mathbf{p}'_{\alpha_\ell}\|^2 + \|\mathbf{p}'_{\beta_\ell}\|^2 + \|\mathbf{p}'_{\gamma_\ell}\|^2 + 2s_\ell^2 \|\mathbf{c}'_{\alpha_\ell}\|^2 - 2s_\ell \|\mathbf{c}'_{\alpha_\ell}\| \|\mathbf{p}'_{\alpha_\ell} - \mathbf{p}'_{\gamma_\ell}\|. \quad (16)$$

This is an expression where the only optimization variable left is s_ℓ . We compute the derivative with respect to s_ℓ and find the root, obtaining

$$4\|\mathbf{c}'_{\alpha_\ell}\|^2 s_\ell - 2\|\mathbf{c}'_{\alpha_\ell}\| \|\mathbf{p}'_{\alpha_\ell} - \mathbf{p}'_{\gamma_\ell}\| = 0. \quad (17)$$

Therefore, the optimal value of s_ℓ , which is positive as required and corresponds to a minimum because the second derivative is positive, is

$$s_\ell = \frac{\|\mathbf{p}'_{\alpha_\ell} - \mathbf{p}'_{\gamma_\ell}\|}{2\|\mathbf{c}'_{\alpha_\ell}\|}. \quad (18)$$

With this, we have already found the optimal parameters (s_ℓ , $\mathbf{R}_\ell$, $\mathbf{t}_\ell$) for the considered problem. Now, notice that using the value of s_ℓ computed in (18), we can obtain the following expression from (15):

$$s_\ell \mathbf{R}_\ell \mathbf{c}'_{\alpha_\ell} = \frac{1}{2} (\mathbf{p}'_{\alpha_\ell} - \mathbf{p}'_{\gamma_\ell}), \quad (19)$$

and similarly, using (12), we get the expressions

$$s_\ell \mathbf{R}_\ell \mathbf{c}'_{\gamma_\ell} = \frac{1}{2} (\mathbf{p}'_{\gamma_\ell} - \mathbf{p}'_{\alpha_\ell}), \quad (20)$$

$$s_\ell \mathbf{R}_\ell \mathbf{c}'_{\beta_\ell} = \mathbf{0}. \quad (21)$$

We can now use the equations (19), (20), and (21) to, finally, provide an expression of E_ℓ in (2) with all parameters having been chosen as optimal. This expression is as follows:

$$E_\ell = \frac{1}{2} \left(\left\| \frac{1}{2} (\mathbf{p}'_{\alpha_\ell} - \mathbf{p}'_{\gamma_\ell}) - \mathbf{p}'_{\alpha_\ell} \right\|^2 + \|\mathbf{p}'_{\beta_\ell}\|^2 + \left\| \frac{1}{2} (\mathbf{p}'_{\gamma_\ell} - \mathbf{p}'_{\alpha_\ell}) - \mathbf{p}'_{\gamma_\ell} \right\|^2 \right). \quad (22)$$

It is noteworthy that this expression depends solely on $\mathbf{p}_{\alpha_\ell}$, $\mathbf{p}_{\beta_\ell}$, and $\mathbf{p}_{\gamma_\ell}$. By operating on this expression, we can formulate the energy E_ℓ in a compact form. For this, let us define $\tilde{\mathbf{p}}_\ell = [\mathbf{p}_{\alpha_\ell}^\top, \mathbf{p}_{\beta_\ell}^\top, \mathbf{p}_{\gamma_\ell}^\top]^\top \in \mathbb{R}^9$. We can then write

$$E_\ell = \frac{1}{2} \tilde{\mathbf{p}}_\ell^\top (\mathbf{B} \otimes \mathbf{I}_3) \tilde{\mathbf{p}}_\ell, \quad (23)$$

where

$$\mathbf{B} = \begin{bmatrix} 1/6 & -1/3 & 1/6 \\ -1/3 & 2/3 & -1/3 \\ 1/6 & -1/3 & 1/6 \end{bmatrix}. \quad (24)$$

Note that since the constraints on the geometry at rest are the same for all the m three-node sets, every set has the same constant matrix $\mathbf{B}$ associated with it. Next, we formulate the deformation energy for the full lattice in (1). Notice that each matrix $\mathbf{Q}_\ell$ is a selector matrix that selects the positions of the nodes (α_ℓ , β_ℓ , and γ_ℓ) of the set $\mathcal{T}_\ell$, in such a way that we have $(\mathbf{Q}_\ell \otimes \mathbf{I}_3) \mathbf{p} = \tilde{\mathbf{p}}_\ell$. With this, we can express (23) as¹

$$\begin{aligned} E_\ell &= \frac{1}{2} \mathbf{p}^\top \left((\mathbf{Q}_\ell \otimes \mathbf{I}_3)^\top (\mathbf{B} \otimes \mathbf{I}_3) (\mathbf{Q}_\ell \otimes \mathbf{I}_3) \right) \mathbf{p} \\ &= \frac{1}{2} \mathbf{p}^\top \left((\mathbf{Q}_\ell^\top \mathbf{B} \mathbf{Q}_\ell) \otimes \mathbf{I}_3 \right) \mathbf{p} \end{aligned} \quad (25)$$

and then the energy for the full lattice is

$$E = \sum_{\ell=1}^m E_\ell = \frac{1}{2} \mathbf{p}^\top \mathbf{A} \mathbf{p}, \text{ with } \mathbf{A} = \left(\sum_{\ell=1}^m \mathbf{Q}_\ell^\top \mathbf{B} \mathbf{Q}_\ell \right) \otimes \mathbf{I}_3, \quad (26)$$

which concludes the proof. $\blacksquare$

B. Deformation Jacobian

From this point onwards, we treat the lattice as a dynamical system represented by its nodes' states at a given time instant. We interpret $E(t)$ as a virtual potential energy associated with the lattice's deformation, as done in [16]. Then, $\mathbf{f}(t) = [\mathbf{f}_1(t)^\top, \mathbf{f}_2(t)^\top, \dots, \mathbf{f}_n(t)^\top]^\top \in \mathbb{R}^{3n}$ denotes the virtual nodal forces associated with $E(t)$, with $\mathbf{f}_\kappa(t) \in \mathbb{R}^3$ being the force at node κ . Observe that $\mathbf{A}$ is a symmetric matrix, because $\mathbf{B}$ is symmetric. Therefore, from (4), the forces have the expression

$$\mathbf{f}(t) = -\nabla_{\mathbf{p}} E(t) = -\mathbf{A} \mathbf{p}(t). \quad (27)$$

Now, similarly to the approaches in [4], [5], [17], [21], [22], [26], we will use a partitioning of the nodes and take differentials (or, more precisely, time derivatives) of (27) to derive a deformation Jacobian. Our partitioning consists in dividing the nodes into two classes: constrained and free. The constrained nodes are those whose positions at every time instant are fixed externally. These are the nodes being gripped by a robotic gripper, or anchored in the environment. The free nodes are those which are not constrained. We denote by n_c the number of constrained nodes, and by n_f the number of free nodes, so that $n_c + n_f = n$. Assuming an arbitrary fixed ordering of the nodes' indices, we arrange the positions of these nodes in vectors $\mathbf{p}_c(t) \in \mathbb{R}^{3n_c}$ and $\mathbf{p}_f(t) \in \mathbb{R}^{3n_f}$, respectively. We arrange $\mathbf{A}$ in blocks accordingly. Then, the time derivative of (27) is

$$-\begin{bmatrix} \mathbf{A}_{cc} & \mathbf{A}_{cf} \\ \mathbf{A}_{fc} & \mathbf{A}_{ff} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{p}}_c(t) \\ \dot{\mathbf{p}}_f(t) \end{bmatrix} = \begin{bmatrix} \dot{\mathbf{f}}_c(t) \\ \dot{\mathbf{f}}_f(t) \end{bmatrix}. \quad (28)$$

¹We use the following properties of the Kronecker product in the final part of the proof: $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{A}\mathbf{B}) \otimes (\mathbf{C}\mathbf{D})$, $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$, and $\mathbf{A} \otimes \mathbf{C} + \mathbf{B} \otimes \mathbf{C} = (\mathbf{A} + \mathbf{B}) \otimes \mathbf{C}$ (see, e.g., [25, Ch. 2]).

Assuming a quasi-static behavior where the object's shape is in equilibrium at all times, the total force at every node κ is always zero. This implies its time derivative is zero too. Notice that this total force is the sum of the ASAP force $\mathbf{f}_\kappa(t)$, associated with the internal deformation, and the total external force on that node. Hence, letting $\mathbf{f}_{e,\kappa}(t)$ denote the external force on node κ , one has $\mathbf{f}_\kappa(t) + \mathbf{f}_{e,\kappa}(t) = [0, 0, 0]^\top$ for $\kappa = 1, 2, \dots, n$. We assume that for the free nodes the external force is zero, or constant (e.g., gravity). Therefore, for every free node κ , one has $\mathbf{f}_{e,\kappa}(t) = [0, 0, 0]^\top$, and $\dot{\mathbf{f}}_\kappa(t) = [0, 0, 0]^\top$. Hence, $\dot{\mathbf{f}}_f(t) = \mathbf{0}$. We use this fact in (28), which results in

$$\mathbf{A}_{fc} \dot{\mathbf{p}}_c(t) + \mathbf{A}_{ff} \dot{\mathbf{p}}_f(t) = \mathbf{0}. \quad (29)$$

It is then possible to obtain directly that

$$\dot{\mathbf{p}}_f(t) = \mathbf{J}_{fc} \dot{\mathbf{p}}_c(t), \text{ with } \mathbf{J}_{fc} = -\mathbf{A}_{ff}^{-1} \mathbf{A}_{fc}. \quad (30)$$

Observe that $\mathbf{J}_{fc} = \partial \mathbf{p}_f / \partial \mathbf{p}_c \in \mathbb{R}^{3n_f \times 3n_c}$ is the deformation Jacobian matrix that relates the differential changes of the positions of the constrained and free nodes of the lattice. Here, $\mathbf{A}_{ff}^{-1}$ is assumed to exist [5], [17]. Notice that, for a given lattice, the matrix $\mathbf{J}_{fc}$ is constant.

III. APPLICATION IN SHAPE CONTROL

ARAP modeling is known to represent the behavior of real elastic deformations well. Our ASAP formulation uses a scaling factor s_ℓ , while ARAP imposes $s_\ell = 1$ for every ℓ . Note that the value of s_ℓ in ASAP is 1 if the three-node set $\mathcal{T}_\ell$ undergoes a rigid transformation (i.e., rotation and translation) with respect to its rest shape, and it remains close to 1 if the set $\mathcal{T}_\ell$ undergoes a non-rigid transformation whose associated deformation is small. This supports the validity of using the derived Jacobian to model elastic objects when the local deformations are small all across the lattice. Moreover, note that, even when all local deformations are small, the global deformation of the lattice can still be large. These considerations lead us to explore the potential use of the proposed Jacobian for shape control of elastic objects under significant deformations. Specifically, our hypothesis is that the proposed Jacobian can model the deformation of this type of objects accurately enough when used together with shape feedback in a control loop. The main advantage in this context is simplicity: the proposed Jacobian is constant and can be obtained offline from the lattice at rest.

We now describe how to use the proposed Jacobian in a typical shape control task. A fixed sensor (e.g., a camera) is assumed to be present in order to provide the shape feedback, such that the positions $\mathbf{p}_\kappa(t)$ for $\kappa = 1, 2, \dots, n$ are expressed in the frame of this sensor. We assume a robot gripper firmly grasps the object and the velocity of the gripper can be set by a controller. One main assumption we need, already stated above, is that the dynamic behavior is quasi-static, i.e., the shape of the object is always in equilibrium at any given instant. In accordance with this, the gripper motions to be employed will not be fast.

In order to have a well-conditioned scenario where the single gripper can exert deformation on the object, we

assume that a region of the object is gripped and a different region is anchored and fixed in the environment. To this end, we divide the constrained nodes in two types: gripped and anchored. Let $\mathbf{p}_g(t) \in \mathbb{R}^{3n_g}$ and $\mathbf{p}_a(t) \in \mathbb{R}^{3n_a}$ denote the vectors containing the positions of gripped and anchored nodes, respectively, such that $n_g + n_a = n_c$. The anchored nodes satisfy $\dot{\mathbf{p}}_a(t) = \mathbf{0}$ at all times. One can define, from (30), matrices $\mathbf{J}_{fg}$ and $\mathbf{J}_{fa}$ taken from the columns of $\mathbf{J}_{fc}$ such that the following expression holds:

$$\dot{\mathbf{p}}_f(t) = \mathbf{J}_{fg} \dot{\mathbf{p}}_g(t) + \mathbf{J}_{fa} \dot{\mathbf{p}}_a(t) = \mathbf{J}_{fg} \dot{\mathbf{p}}_g(t). \quad (31)$$

Frequently, for simplicity or due to task requirements (e.g., assembling some specific part of an object on another object), one wants to control explicitly only some region of the object, i.e., only a subset of the free nodes. For this, let us define a partition of the free nodes in n_s servoed nodes and n_u unservoed nodes, with $n_s + n_u = n_f$. This is done in the same way as above: we define $\mathbf{p}_s(t) \in \mathbb{R}^{3n_s}$ and $\mathbf{p}_u(t) \in \mathbb{R}^{3n_u}$ as the vectors containing respectively the positions of servoed and unservoed nodes. We can obtain a Jacobian $\mathbf{J}_{sg}$ by selecting the rows of $\mathbf{J}_{fg}$ corresponding to servoed nodes, such that

$$\dot{\mathbf{p}}_s(t) = \mathbf{J}_{sg} \dot{\mathbf{p}}_g(t). \quad (32)$$

We assume the gripped nodes are rigidly coupled to the gripper and we represent the gripper's configuration by a Cartesian frame. The pose of the gripper is denoted by $\mathbf{x}_h(t) \in \mathbb{R}^6$ and $\mathbf{v}_h(t) \in \mathbb{R}^6$ denotes its velocity vector, comprising its linear and angular velocity in 3D expressed in the sensor frame. As in [5], [17], one can define a mapping

$$\dot{\mathbf{p}}_g(t) = \mathbf{G}_{gh}(t) \mathbf{v}_h(t), \quad (33)$$

where $\mathbf{G}_{gh}(t)$ can be computed from the gripper's time-varying pose. Then, using (32), one can write

$$\dot{\mathbf{p}}_s(t) = \mathbf{J}_{sh}(t) \mathbf{v}_h(t), \quad (34)$$

where $\mathbf{J}_{sh}(t) = \mathbf{J}_{sg} \mathbf{G}_{gh}(t)$. Now, let us assume that the goal of the control task is to drive the servoed nodes toward some constant desired positions, $\mathbf{p}_s^*$.

We propose a linear adaptive MPC exploiting the Jacobian $\mathbf{J}_{sh}(t)$. The goal is to find an optimal control sequence $\mathcal{V}_h = (\mathbf{v}_h(0|t), \mathbf{v}_h(1|t), \dots, \mathbf{v}_h(N-1|t))$ that minimizes the following quadratic function, defined over a prediction horizon N :

$$\min_{\mathcal{V}_h} \sum_{k=0}^{N-1} \left(\|\mathbf{p}_s(k+1|t) - \mathbf{p}_s^*\|_{\mathbf{Q}}^2 + \|\mathbf{v}_h(k|t)\|_{\mathbf{R}}^2 + \|\mathbf{x}_h(k+1|t) - \mathbf{x}_h^*\|_{\mathbf{S}}^2 \right). \quad (35)$$

Here, the notation $\mathbf{z}(k|t)$ represents the predicted value of a variable k steps ahead of the current real-time instant t . The operator $\|\mathbf{z}\|_{\mathbf{W}}^2 = \mathbf{z}^\top \mathbf{W} \mathbf{z}$ denotes the weighted squared Euclidean norm. The matrices $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{S}$ are positive-definite weighting matrices that penalize the control error, the control effort, and the deviation of the gripper's pose $\mathbf{x}_h$ from a nominal $\mathbf{x}_h^*$, respectively.

The minimization problem is subject to the initial conditions, the linearized system dynamics, and the physical limitations. Given the control interval T , the constraints are defined for $k = 0, \dots, N-1$ as follows:

$$\mathbf{p}_s(0|t) = \mathbf{p}_s(t), \quad (36a)$$

$$\mathbf{x}_h(0|t) = \mathbf{x}_h(t), \quad (36b)$$

$$\mathbf{p}_s(k+1|t) = \mathbf{p}_s(k|t) + T \mathbf{J}_{sh}(t) \mathbf{v}_h(k|t), \quad (36c)$$

$$\mathbf{x}_h(k+1|t) = \mathbf{x}_h(k|t) + T \mathbf{v}_h(k|t), \quad (36d)$$

$$\mathbf{v}_h^{\min} \leq \mathbf{v}_h(k|t) \leq \mathbf{v}_h^{\max}, \quad (36e)$$

$$\Delta \mathbf{x}_h^{\min} \leq \mathbf{x}_h(k+1|t) - \mathbf{x}_h^* \leq \Delta \mathbf{x}_h^{\max}, \quad (36f)$$

where the vectors $\mathbf{v}_h^{\min}, \mathbf{v}_h^{\max} \in \mathbb{R}^6$ represent the velocity limits of the gripper, while $\Delta \mathbf{x}_h^{\min}, \Delta \mathbf{x}_h^{\max} \in \mathbb{R}^6$ represent the deviation limits of the gripper pose from a reference.

We approximate the evolution of the pose in (36d) by assuming that the geometric angular velocity is equivalent to the time-derivative of the rotation vector. Although this only works for small velocities, the cumulative integration error is mitigated because the predicted $\mathbf{x}_h(0|t)$ is re-initialized with the current real pose $\mathbf{x}_h(t)$ at each control step. Similarly, the Jacobian $\mathbf{J}_{sh}(t)$ is evaluated using the currently measured pose at time t and held constant across the prediction horizon. This approximation allows the optimization at each control step to be formulated as an efficient convex Quadratic Program (QP).

Finally, the current input $\mathbf{v}_h$ is selected in each control step as follows:

$$\mathbf{v}_h(t) = \mathbf{v}_h(0|t). \quad (37)$$

IV. SIMULATION RESULTS

This section presents simulation results in multiple scenarios to evaluate the effectiveness of the lattice model for control applications. Supplementary videos are available at: <https://doi.org/10.5281/zenodo.19352191>.

Two deformable objects are considered: a 2D sheet-like lattice and a 3D beam-like lattice. The rest shapes of these lattices, as well as the designated servoed, gripped and anchored nodes, are depicted in Fig. 1. In both cases, the primary control objective is to move the servoed nodes of the deformable lattice to their desired positions, $\mathbf{p}_s^*$. We choose the number ($n_s = 3$) and distribution of the servoed nodes so that the evaluated scenarios represent non-trivial tasks requiring refined control on the overall shape of the lattice.

The control strategy is the linear adaptive MPC approach described in Section III, with the following parameters (all in SI units):

$$\begin{aligned} T &= 0.02, & N &= 10, \\ \mathbf{v}_h^{\max} &= [0.05 \cdot \mathbf{1}_3^\top, 0.1 \cdot \mathbf{1}_3^\top]^\top, & \mathbf{v}_h^{\min} &= -\mathbf{v}_h^{\max}, \\ \Delta \mathbf{x}_h^{\max} &= [\mathbf{1}_3^\top, 0.5 \pi \cdot \mathbf{1}_3^\top]^\top, & \Delta \mathbf{x}_h^{\min} &= -\Delta \mathbf{x}_h^{\max}, \\ \mathbf{x}_h^* &= \mathbf{x}_h(0), & \mathbf{Q} &= 0.001^{-2} \cdot \mathbf{I}_{3n_s}, \\ \mathbf{R} &= \text{diag}(\mathbf{v}_h^{\max})^{-2}, & \mathbf{S} &= \text{diag}(\Delta \mathbf{x}_h^{\max})^{-2}, \end{aligned} \quad (38)$$

where $\mathbf{1}_3 = [1, 1, 1]^\top$ and $\text{diag}(\cdot)$ denotes a diagonal matrix having the values in parentheses in the diagonal. While the

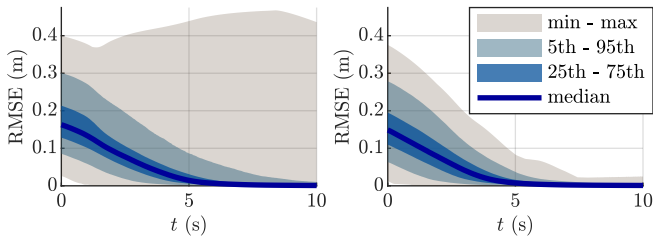


Fig. 2. Evolution of the RMSE over time for the 2D (left) and 3D (right) lattices, derived from 10000 independent simulations. The graphs show the 25-75th percentile range, the 5-95th percentile range, the minimum-maximum range, and the median.

controller uses the proposed ASAP lattice model to compute the system Jacobian $\mathbf{J}_{sh}(t)$, the ground-truth deformations of the lattices are simulated using the state-of-the-art ARAP model [16], [19].

All computations are performed in MATLAB on an Intel i7-10700 CPU, using the Optimization Toolbox's quadprog solver for the QP. The controller executes in an average of 2.5 ms per time step.

A. Monte Carlo Analyses

Monte Carlo simulations were conducted, comprising 10000 independent trials for each object. In each trial, a randomized gripper pose displacement is sampled from a uniform probability distribution to generate a feasible target shape. The distribution bounds for each Cartesian coordinate are ± 0.2 m for translational displacements and ± 0.15 rad for rotational displacements. The lattice is deformed based on the randomized gripper displacement. The resulting servoed node positions, $\mathbf{p}_s$, are set as the desired reference positions, $\mathbf{p}_s^*$, for the control in that trial. Subsequently, the system is reset to its rest configuration, and the control algorithm is executed to drive the servoed nodes toward $\mathbf{p}_s^*$. The metric chosen to evaluate the control performance is the root mean square error of the servoed nodes, $\text{RMSE} = \|\mathbf{p}_s - \mathbf{p}_s^*\| / \sqrt{n_s}$.

The results are summarized in Fig. 2. On one hand, the 95th and lower percentiles show that the control strategy works well and performs very similarly for both lattices. On the other hand, for the 2D lattice, the maximum values show that in some outlier trials there is a very poor control performance, at least in the evaluated time window. In contrast, for all the 3D lattice trials, there is convergence to small control errors. For a qualitative visual assessment, the top row of Fig. 3 depicts the resulting 3D lattice shapes of the 2500th, 1000th and 1st worst trials at $t = 10$ s. In all these instances, the positions of the servoed nodes closely match the desired ones. On the other hand, the bottom row of Fig. 3 shows the worst-case trial of the 2D lattice. Specifically, the lattice shape at $t = 10$ s (left) shows a large discrepancy between the current and desired positions of the servoed nodes. However, in this case, the control manages to converge at $t = 24$ s (right). Therefore, the control objective is ultimately achieved, albeit with a much slower convergence. This challenging case exemplifies the limitations of the proposed Jacobian under certain complex

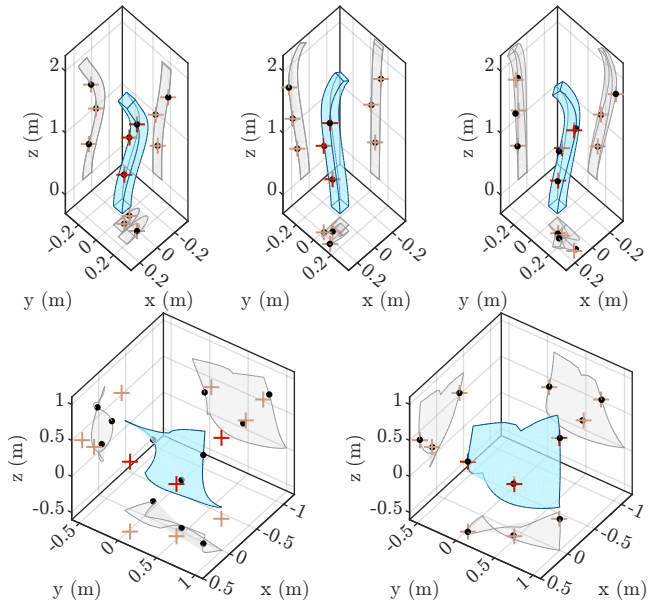


Fig. 3. Examples of the manipulated lattices. The servoed nodes are depicted as black dots, while their desired positions are depicted as red crosses. Top row: 3D lattice configurations of the 2500th (left), 1000th (middle) and 1st (right) worst trials at $t = 10$ s. Bottom row: 2D lattice configuration of the worst trial at $t = 10$ s (left), and the resulting converged shape at $t = 24$ s (right).

highly non-linear deformation behaviors of the ARAP model.

B. Additional Analyses

The simulations presented thus far have used the standard ARAP model. While this geometric framework eliminates the need for physical material parameters, it cannot inherently account for gravitational effects, which affects highly flexible structures like cloth. To evaluate the proposed controller under these conditions, we add a downward vector weighted by a constant parameter λ for the free nodes in the system of equations solved in ARAP [16, Eq. (9)]. This is a simple and plausible way to represent the effect of gravity.

We applied the worst-case scenario from the 2D Monte Carlo simulations (depicted in Fig. 3) and repeated the simulation with gravity constants $\lambda = 0.002$ and $\lambda = 0.005$. Following the same open-loop initialization, new feasible desired positions $\mathbf{p}_s^*$ were generated. The system was then reset to its rest configuration before executing the closed-loop control. As shown in Fig. 4, the inclusion of gravity significantly improves control performance: the servoed nodes' positions at $t = 10$ s closely match the new $\mathbf{p}_s^*$.

Finally, the controller's robustness against measurement noise is analyzed. Using the reference $\mathbf{p}_s^*$ from the 2500th worst trial of the 3D lattice (Fig. 3, top left), we added uncorrelated zero-mean Gaussian white noise to the perceived servoed node positions $\mathbf{p}_s$ that are used by the controller. The test was conducted using three different standard deviations, σ . As illustrated in Fig. 5, the transient response and the final error worsen as the standard deviation increases. Nevertheless, the control achieves acceptable results even with a standard deviation of up to $\sigma = 0.1$ m.

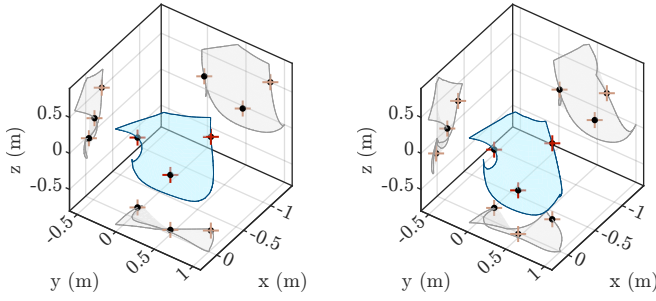


Fig. 4. Examples of the manipulated lattice for the previously identified 2D worst-case trial but repeated with simulated gravity. Lattice configurations at $t = 10$ s for gravity constants $\lambda = 0.002$ (left) and $\lambda = 0.005$ (right). The servoed nodes are depicted as black dots, while their desired positions are depicted as red crosses.

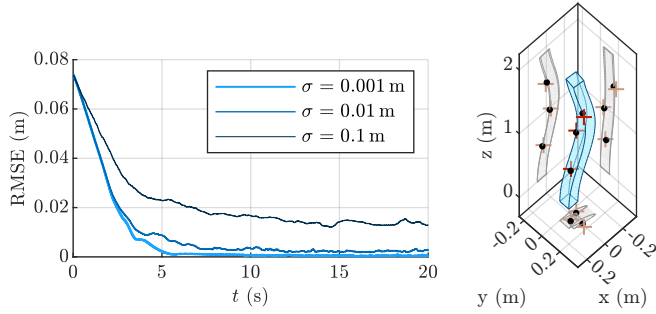


Fig. 5. Results with Gaussian noise added to the measured servoed positions. Left: evolution of the RMSE for varying noise levels. Right: resulting configuration for $t = 20$ s and the maximum tested noise level, $\sigma = 0.1$ m.

V. CONCLUSION

This paper has presented a simple lattice-based deformation Jacobian model, validating its potential applicability in shape control tasks in a simulated setup. In future work, we wish to conduct further analysis and evaluations of this model. Our ultimate goal is to apply the model on real-world deformable objects for tasks such as assembly, folding, or transport. In this context, interesting extensions of the proposed approach include the control with time-varying shape references and the handling of obstacles and contacts in the environment.

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